`GeoModels` Tutorial: simulation, estimation and prediction of global spatial data on the planet Earth using Gaussian random fields.

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Introduction

This tutorial illustrates how to analyze spherical data (*i.e.*, spatial data on the surface of a sphere) with the R package `GeoModels` (Bevilacqua et al., 2026). Albeit the main focus of the tutorial is on Gaussian random fields (GRFs) defined over a sphere of arbitrary radius, non-Gaussian random fields implemented in `GeoModels` can also be coupled with spherical data.

We denote with S^2 the sphere of $\mathbb{R}^3$ of arbitrary radius $R > 0$, defined as

$$S^2 = \{\boldsymbol{x} \in \mathbb{R}^3 : \|\boldsymbol{x}\| = R\}.$$

This tutorial shows how to simulate, estimate and predict GRFs on a sphere of arbitrary radius using a specific covariance model proposed in Alegría et al. (2021), which is quite flexible for the reasons explained below. We start by loading the libraries needed for the analysis and set the radius of the sphere:

```
rm(list=ls())
library(GeoModels)
library(mapproj)
library(maps)
library(globe)
library(fields)
library(sphereplot)
set.seed(1891)
```

Gaussian random fields defined on the sphere

We consider a zero-mean and unit-variance GRF, $Y = \{Y(\boldsymbol{x}), \boldsymbol{x} \in S^2\}$, defined on a sphere of radius $R > 0$.

The great-circle (GC) distance corresponds to the shortest path joining two points on the spherical surface and is the natural distance to be used when analyzing data on a sphere. GC distance between two points $\boldsymbol{x}_i, \boldsymbol{x}_j \in S^2$ can be computed as

$$d_{GC}(\boldsymbol{x}_i, \boldsymbol{x}_j) = R \arccos\left(\frac{\boldsymbol{x}_i^T \boldsymbol{x}_j}{R^2}\right).$$

Using the geometric definition of the dot product $\boldsymbol{x}_i^T \boldsymbol{x}_j = \|\boldsymbol{x}_i\|\|\boldsymbol{x}_j\| \cos(\theta_{ij}) = R^2 \cos(\theta_{ij})$,

this implies that GC distance can be computed as $d_{GC}(\boldsymbol{x}_i, \boldsymbol{x}_j) = R\theta_{ij}$, where θ_{ij} is the angle between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$.

A point $\boldsymbol{x} \in S^2$ can be parametrized through spherical coordinates using radians $(\text{lon}^*, \text{lat}^*, R)$, where $\text{lon}^* \in [-\pi, \pi]$ is the longitude and $\text{lat}^* \in [-\pi/2, \pi/2]$ is the latitude. The connection between Euclidean and spherical coordinates is given by $\boldsymbol{x} = (x_1, x_2, x_3)^T$, with

$$x_1 = R\cos(\text{lon}^*)\cos(\text{lat}^*), \quad x_2 = R\sin(\text{lon}^*)\cos(\text{lat}^*), \quad x_3 = R\sin(\text{lat}^*).$$

Alternatively, one can use decimal degrees $(\text{lon}, \text{lat}, R) = (180\text{lon}^*/\pi, 180\text{lat}^*/\pi, R)$, where $\text{lon} \in [-180, 180]$ and $\text{lat} \in [-90, 90]$ (the `GeoModels` package uses decimal degrees).

To find the angle θ_{ij} between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, we use the relation $\theta_{ij} = \arccos(\boldsymbol{x}_i^T\boldsymbol{x}_j/R^2)$; using spherical coordinates, we obtain

$$\theta_{ij} = \arccos\{\sin(\text{lat}_i^*)\sin(\text{lat}_j^*) + \cos(\text{lat}_i^*)\cos(\text{lat}_j^*)\cos(\text{lon}_i^* - \text{lon}_j^*)\}.$$

Alternatively, one can compute the chordal (CH) distance, that is, the segment below the arc joining any pair of points located on the spherical shell. It is given by

$$d_{CH}(\boldsymbol{x}_i, \boldsymbol{x}_j) = 2R\sin(\theta_{ij}/2). \tag{1}$$

CH underestimates the GC distance, and the approximation error increases with the size of the considered portion of the planet.

We assume $\text{cor}(Y(\boldsymbol{x}_i), Y(\boldsymbol{x}_j)) = \rho(d_{GC}(\boldsymbol{x}_i, \boldsymbol{x}_j))$, where $\rho : [0, R\pi] \to \mathbb{R}$ is continuous with $\rho(0) = 1$. Under this setting, the correlation function is called *geodesically isotropic* (Gneiting, 2013; Porcu et al., 2016). We then consider the location and scale transformation

$$Z(\boldsymbol{x}) = \mu(\boldsymbol{x}) + \sigma Y(\boldsymbol{x}),$$

where $\mu(\boldsymbol{x}) \in \mathbb{R}$ is a spherical varying mean and $\sigma > 0$. The package `GeoModels` allows a mean-regression specification, that is, $\mu(\boldsymbol{x}) = M(\boldsymbol{x})^T\boldsymbol{\beta}$, where $M(\boldsymbol{x})$ is a vector of spherical covariates. Nevertheless, for simplicity, this tutorial assumes a constant mean: $\mu(\boldsymbol{x}) = \mu$.

Then $\text{E}(Z(\boldsymbol{x})) = \mu$ and $\text{cov}(Z(\boldsymbol{x}_i), Z(\boldsymbol{x}_j)) = \sigma^2\rho(d_{GC}(\boldsymbol{x}_i, \boldsymbol{x}_j))$, where σ^2 is the variance parameter. In order to obtain a realization from a spherical RF, we need to specify a mean, a variance parameter and a valid (*i.e.*, positive-definite) geodesically isotropic parametric correlation function $\rho_\gamma(\cdot)$. We work on a rescaled representation of the Earth with sphere radius $R = 1$ hereafter:

```
radius=1
```

We set the spherical coordinates in decimal degrees on the unit sphere using the function `pointsphere` of the package `sphereplot` (Robotham, 2013):

```
NN=700 ## number of location sites on the sphere
coords=pointsphere(NN,c(-180,180),c(-90,90),
                   c(radius,radius))[,1:2]
```

We can visualize (Figure 1) the locations on the planet Earth using some functions of the package `globe` (Baddeley et al., 2017) with the following code:

```
globeearth(eye=place("newyorkcity"))
globepoints(loc=coords,pch=20,cex=0.4)
globeearth(eye=place("everest"))
globepoints(loc=coords,pch=20,cex=0.4)
```



Figure 1: Location sites on the planet Earth used in this tutorial

An issue when considering correlation models defined on the sphere is that classical models defined on Euclidean spaces are not necessarily suitable on the sphere (Gneiting, 2013). For instance, the popular Matérn model equipped with the great-circle distance is defined as

$$M_{\nu,\alpha}(d_{GC}) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{d_{GC}}{\alpha}\right)^{\nu} K_{\nu}\left(\frac{d_{GC}}{\alpha}\right),$$

and it is not valid on the sphere when $\nu > 0.5$. Here K_{ν} is a modified Bessel function of the second kind of order ν.

An interesting covariance model defined on the unit sphere having the same features of the Matérn model is the so-called $\mathcal{F}$ family of correlation functions proposed in Alegría

et al. (2021), defined as

$$\mathcal{F}_{\tau,\alpha,\nu}(d_{GC}) = \frac{B(\alpha,\nu+\tau)}{B(\alpha,\nu)}\,{}_2F_1(\tau,\alpha,\alpha+\nu+\tau;\cos(d_{GC})), \quad 0 \le d_{GC} \le \pi, \tag{2}$$

where τ, α and ν are strictly positive parameters and where ${}_2F_1$ is the Gauss hypergeometric function

$${}_2F_1(a,b,c;x) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k} \frac{x^k}{k!}, \qquad |x| < 1,$$

with $(\cdot)_k$ being the Pochhammer symbol. Finally, $B(\cdot,\cdot)$ is the Beta function. Alegría et al. (2021) show that (2) is k times differentiable at the origin if and only if $\lfloor \nu/2 \rfloor > k$, for any τ, α, where $\lfloor x \rfloor$ denotes the greatest integer not exceeding $x \in \mathbb{R}$. From this point of view, the $\mathcal{F}$ class of correlation functions can be viewed as the analogue of the Matérn class on the sphere. In the `GeoModels` package the $\mathcal{F}$ family is implemented as

$$\begin{aligned}\mathcal{F}_{1/\alpha,1/\alpha+0.5,\nu}(d_{GC}) = &\frac{B(1/\alpha+0.5,\nu+1/\alpha)}{B(1/\alpha+0.5,\nu)} \\ &\times {}_2F_1(1/\alpha,1/\alpha+0.5,2/\alpha+0.5+\nu;\cos(d_{GC})).\end{aligned} \tag{3}$$

This parametrization addresses identifiability issues associated with (2); the α and ν parameters are correlation range and smoothness parameters, respectively, as in the Matérn model.

We set the $\mathcal{F}$ correlation model with associated parameters and the other parameters of the GRF. In `GeoModels` the $\mathcal{F}$ model is implemented under the name `F_Sphere`:

```
corrmodel="F_Sphere" ## correlation model and parameters
scale=0.2
smooth=0.5
sill=1
nugget=0
mean=0
```

Here `scale` corresponds to α, `sill` is the variance parameter σ^2, and `smooth` is the smoothness parameter ν. The choice $\nu = 0.5$ leads to a non-mean-square-differentiable Gaussian random field. We are now ready to simulate a GRF on the unit sphere using `GeoSim`:

```
param=list(mean=mean,sill=sill,nugget=nugget,
          smooth=smooth,scale=scale)
data=GeoSim(coordx=coords,corrmodel=corrmodel,
           param=param,distance="Geod",radius=radius)$data
```

Note that `distance="Geod"`, coupled with location sites in lon/lat format in decimal degrees (`coordx=coords`) and a valid correlation model on the sphere (`corrmodel`) with a given radius, allows simulation based on Cholesky decomposition of a GRF on S^2.

Estimation of Gaussian random fields on the sphere

Let $\rho_{\alpha,\nu}(d_{GC}) = \mathcal{F}_{1/\alpha,1/\alpha+0.5,\nu}(d_{GC})$. Given a realization $\boldsymbol{Z} = (Z(\boldsymbol{x}_1), Z(\boldsymbol{x}_2), \ldots, Z(\boldsymbol{x}_N))^T$ from a GRF defined on S^2 with correlation (3), we can perform maximum-likelihood estimation by maximizing the log-likelihood function

$$\begin{aligned}\ell(\boldsymbol{\beta}) = & -\frac{1}{2}\log|\sigma^2 C_{\alpha,\nu}| \\ & -\frac{1}{2\sigma^2}(\boldsymbol{Z}-\mu\mathbf{1})^T C_{\alpha,\nu}^{-1}(\boldsymbol{Z}-\mu\mathbf{1}),\end{aligned}$$

up to an additive constant, with respect to $\boldsymbol{\beta} = (\mu, \sigma^2, \alpha, \nu)^T$. The correlation matrix is

$$C_{\alpha,\nu} = [\rho_{\alpha,\nu}\{d_{GC}(\boldsymbol{x}_i, \boldsymbol{x}_j)\}]_{i,j=1}^N .$$

We can use `GeoFit` to perform maximum-likelihood estimation. We first choose the fixed parameters and the parameters that must be estimated:

```
fixed<-list(nugget=nugget)
start<-list(mean=mean,scale=scale,sill=sill,smooth=smooth)
```

We consider a quasi-Newton optimization as implemented in the `nlminb` function, which allows box-constrained optimization using PORT routines. We first set lower and upper bounds:

```
optimizer="nlminb"
I=Inf
upper<-list(mean=I,scale=I,sill=I,smooth=I)
lower<-list(mean=-I,scale=1e-6,sill=1e-6,smooth=1e-6)
```

We are now ready to perform maximum-likelihood estimation using `GeoFit`. This option can be time consuming depending on the number of points on the sphere:

```
fit_geo_ml<-GeoFit(data=data,coordx=coords,
          corrmodel=corrmodel,likelihood="Full",type="Standard",
          optimizer=optimizer,lower=lower,upper=upper,
          start=start,fixed=fixed,distance="Geod",radius=radius)
```

The default maximization algorithm used in `GeoFit` is the Nelder–Mead method as implemented in `optim`; here we explicitly select `nlminb`.

Since maximum likelihood can be computationally expensive, particularly when computing the correlation function involves special functions, as in the Matérn or $\mathcal{F}$ cases, we can use conditional pairwise-likelihood estimation (Bevilacqua and Gaetan, 2015), obtained by maximizing

$$pl(\boldsymbol{\beta}) = \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} w_{ij} \left[\log f_{Z_i, Z_j}(z_i, z_j; \boldsymbol{\beta}) - \log f_{Z_j}(z_j; \boldsymbol{\beta}) \right]. \tag{4}$$

It can be performed with the following code:

```
fit_geo_pl<-GeoModels::GeoFit(data=data,coordx=coords,
          corrmodel=corrmodel,likelihood="Conditional",
          type="Pairwise",neighb=3,optimizer=optimizer,
          lower=lower,upper=upper,start=start,fixed=fixed,
          distance="Geod",radius=radius)
```

In (4), $f_{Z_i, Z_j}(z_i, z_j; \boldsymbol{\beta})$ and $f_{Z_j}(z_j; \boldsymbol{\beta})$ are Gaussian bivariate and marginal densities, respectively, and w_{ij} are non-negative weights not depending on $\boldsymbol{\beta}$, specified as

$$w_{ij}(m) = \begin{cases} 1, & \boldsymbol{x}_i \in N_m(\boldsymbol{x}_j), \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

where $N_m(\boldsymbol{x}_j)$ is the set of the m nearest neighbours of the site $\boldsymbol{x}_j$. In this example $m = 3$.

The two estimates can be obtained from the objects `fit_geo_ml` and `fit_geo_pl`:

```
unlist(fit_geo_ml$param)
unlist(fit_geo_pl$param)
```

The original tutorial reported the following estimates (which may differ across versions of the software):

```
# Full likelihood: mean, scale, sill, smooth
# -0.0768565 0.1669835 0.9300575 0.5530937
# Conditional pairs: mean, scale, sill, smooth
# -0.08115149 0.21449024 0.93404028 0.46148253
```

Recall that the Matérn model is not, in general, a suitable covariance model on the sphere when coupled with geodesic distance, but it is a valid model when coupled with chordal

distance. In GeoModels, estimation can also be performed using CH distance or Euclidean distance after a suitable projection of the spherical coordinates. Computation of CH distances involved in the covariance matrix can be obtained using (1).

We can use conditional pairwise-likelihood estimation with chordal distances (setting distance="Chor") to estimate a Matérn covariance model:

```
fit_chor_pl<-GeoModels::GeoFit(data=data,coordx=coords,
   corrmodel="Matern",likelihood="Conditional",type="Pairwise",
   neighb=3,optimizer=optimizer,lower=lower,upper=upper,
   start=start,fixed=fixed,distance="Chor",radius=radius)
```

We can also consider conditional pairwise-likelihood estimation of the Matérn model using a suitable projection of the spherical points coupled with Euclidean distance. We use the function mapproject from the package mapproj (McIlroy et al., 2017) to obtain projected coordinates using a sinusoidal projection:

```
orientation=c(90,0,0)
prj=mapproject(coords[,1],coords[,2],projection="sinusoidal",
              orientation=orientation)
coords_prj=radius*cbind(prj$x,prj$y)
```

We can visualize the projected points (Figure 2) with the following code. The map and the locations must use the same projection orientation:

```
sinusoidal.proj=map(database="world",ylim=c(-90,90),
 xlim=c(-180,180),col="grey80",fill=TRUE,plot=FALSE,
 projection="sinusoidal",orientation=orientation)
maps::map(sinusoidal.proj)
points(coords_prj,pch=20,cex=0.4)
```

Other types of projections can be chosen in mapproj, such as Mercator or cylindrical projections.

Then we can estimate with conditional pairwise likelihood using Euclidean distances:

```
fit_eucl_pl<-GeoFit(data=data,coordx=coords_prj,
   corrmodel="Matern",neighb=3,likelihood="Conditional",
   type="Pairwise",optimizer=optimizer,lower=lower,
   upper=upper,start=start,fixed=fixed,distance="Eucl",
   radius=radius)
```

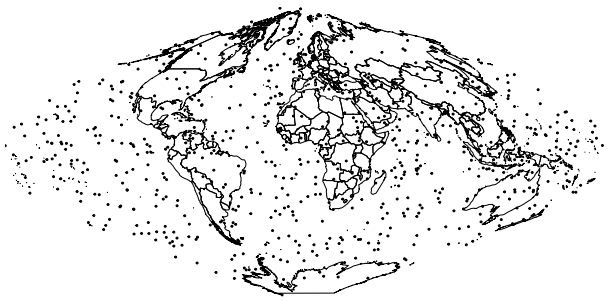


Figure 2: Sinusoidal projection

The two estimates of the Matérn model using chordal and Euclidean distances can be obtained from the objects `fit_chor_pl` and `fit_eucl_pl` as:

```
unlist(fit_chor_pl$param)
unlist(fit_eucl_pl$param)
```

In the original tutorial, the estimates (mean, scale, sill, smooth) were:

```
# Chordal:
# -0.08149043 0.16274464 0.93417666 0.44757851
# Projected Euclidean:
# -0.08055844 0.14207861 0.92881109 0.42005100
```

These values should be checked by running the R script in the current version of `GeoModels`.

Boxplots of the distances (Figure 4) can be useful to depict the different behaviour of these three types of distances in this specific example:

```
geod_ds=rdist.earth(coords,miles=FALSE,R=radius)
chor_ds=2*radius*sin(geod_ds/(2*radius))
eucl_ds=as.matrix(dist(coords_prj))
pairs=upper.tri(geod_ds)
boxplot(geod_ds[pairs],chor_ds[pairs],eucl_ds[pairs],
```

```
      names=c("Greatcircle","Chordal","Euclidean"))
```

Given the estimation of the mean and variance parameters, the standardized residuals

$$\widehat{Y}(\boldsymbol{x}_i) = \frac{Z(\boldsymbol{x}_i) - \widehat{\mu}}{\sqrt{\widehat{\sigma}^2}}, \qquad i = 1, \ldots, N,$$

can be viewed as a realization of a zero-mean unit-variance spherical GRF with correlation model (3). Using conditional pairwise-likelihood estimates, the residuals can be computed using `GeoResiduals`:

```
res=GeoResiduals(fit_geo_pl)
```

Then the marginal distribution assumption on the residuals can be graphically checked with a QQ-plot (Figure 3, left) using `GeoQQ`:

```
### checking model assumptions: marginal distribution
GeoQQ(res)
```

The correlation model assumption can be graphically checked by comparing the empirical and the estimated semivariogram functions using `GeoVariogram` and `GeoCovariogram` (Figure 3, right):

```
### checking model assumptions: variogram model
vario=GeoVariogram(data=res$data,coordx=coords,
  maxdist=max(geod_ds)/2,distance="Geod",radius=radius)
GeoCovariogram(res,vario=vario,show.vario=TRUE,pch=20,
              ylim=c(0,1.2))
```

Prediction of Gaussian random fields on the sphere

For a given location $\boldsymbol{x}_0 \in S^2$, the optimal prediction in this example is computed by

$$\widehat{Z}(\boldsymbol{x}_0) = \widehat{\mu} + \sum_{i=1}^{N} \widehat{\lambda}_i [Z(\boldsymbol{x}_i) - \widehat{\mu}], \tag{6}$$

where $\widehat{\mu}$ is a mean estimate and the vector of weights $\widehat{\boldsymbol{\lambda}} = (\widehat{\lambda}_1, \ldots, \widehat{\lambda}_N)^T$ is given by $\widehat{\boldsymbol{\lambda}} = \widehat{C}^{-1}_{\widehat{\alpha},\widehat{\nu}} \widehat{\boldsymbol{c}}_{\widehat{\alpha},\widehat{\nu}}$, where $\widehat{C}_{\widehat{\alpha},\widehat{\nu}}$ is the estimated correlation matrix and $\widehat{\boldsymbol{c}}_{\widehat{\alpha},\widehat{\nu}}$ is the estimated correlation vector between the observed locations and the point to predict. Moreover, the associated mean squared error is computed through

$$\mathrm{MSE}(\widehat{Z}(\boldsymbol{x}_0)) = \widehat{\sigma}^2 \left(1 - \widehat{\boldsymbol{c}}^T_{\widehat{\alpha},\widehat{\nu}} \widehat{C}^{-1}_{\widehat{\alpha},\widehat{\nu}} \widehat{\boldsymbol{c}}_{\widehat{\alpha},\widehat{\nu}}\right). \tag{7}$$

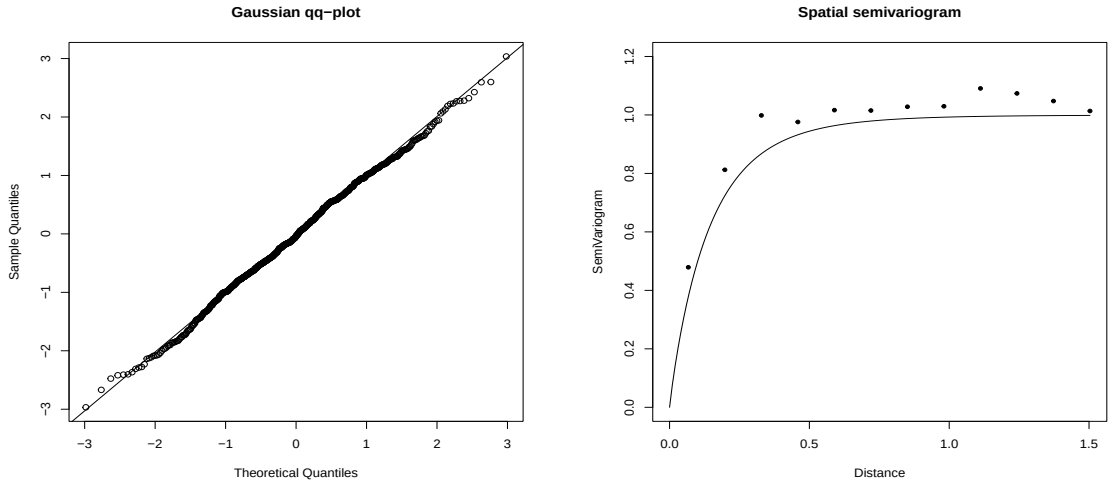


Figure 3: Left: QQ-plot for the model residuals. Right: empirical vs estimated semivariogram function for the residuals

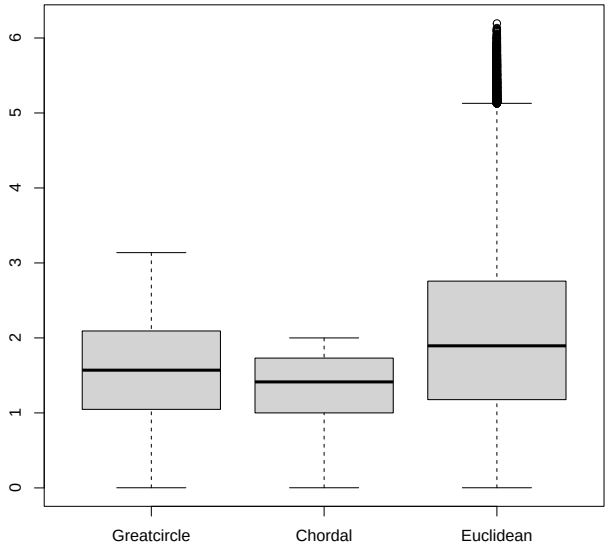


Figure 4: Boxplot of great-circle, chordal and Euclidean distances (from left to right)

These formulas correspond to simple kriging with estimated parameters treated as fixed. The tutorial uses a local version with 100 neighbours, so the covariance matrices and vectors in (6)–(7) are restricted to the selected neighbourhood.

Optimal local prediction and associated MSE can be computed using `GeoKrigloc`. Suppose we want to predict the GRF around the north-pole area. First we specify the spherical locations to predict (in lon/lat format):

```
## locations to predict given in lon/lat format
loc_to_pred=as.matrix(expand.grid(seq(-180,180,2.5),
                                  seq(45,90,0.15)))
```

To speed up computation of (6) and (7), the optimal local predictor using 100 neighbours and its associated MSE (using the maximum conditional pairwise-likelihood estimates) can be obtained with

```
pr_geo=GeoKrigloc(fit_geo_pl,loc=loc_to_pred,
                  mse=TRUE,neighb=100)
```

The matrix of locations to predict, predictions and associated MSE can be obtained as

```
predictions=cbind(loc_to_pred,pr_geo$pred,pr_geo$mse)
head(predictions)
```

The original tutorial reported sample values of these predictions; the values should be refreshed after running the code in the current version of `GeoModels`.

A graphical visualization (Figure 5) of the prediction and associated MSE can be obtained with

```
globeearth(eye=place("newyorkcity"))
globepoints(loc=loc_to_pred,pch=20,
  col=terrain.colors(length(pr_geo$pred),alpha=0.03)[rank(pr_geo$pred)])
globeearth(eye=place("newyorkcity"))
globepoints(loc=loc_to_pred,pch=20,
  col=cm.colors(length(pr_geo$mse),alpha=0.03)[rank(pr_geo$mse)])
```

References

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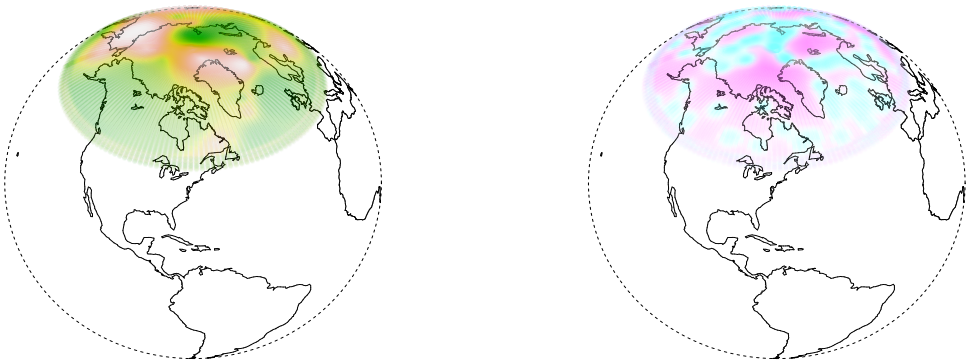


Figure 5: From left to right: prediction and associated MSE in the north-pole area

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